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The Effect of Cracks on the Compressibility of Rock[1]

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Abstract. Compressibility of porous material is greater than that of solid material of the same composition, and the difference is shown to be equal to rate of change of porosity with pressure, for any pore shape or concentration. Expressions for compressibility are given for two special cases for material of low pore concentration: for spherical pores and for narrow cracks. Comparison of the two cases shows that a crack increases compressibility nearly as much as a spherical pore of the same diameter as the length of the crack, although porosity in these two cases differs enormously. For material in which all porosity occurs as narrow cracks, it is shown that porosity can, in certain cases, be determined quite precisely from compressibility measurements.

INTRODUCTION

Measurement of the deformation of rocks under hydrostatic pressure shows that compressibility—the fractional volume decrease per unit of pressure—often is not a constant, as would be expected for an elastic material, but varies with pressure and with the type of test. For example, *Adams and Williamson* [1923] and *Zisman* [1933] found that compressibility of samples jacketed with material impervious to the pressurizing medium usually decreases markedly with increasing pressure, whereas compressibility of unjacketed rocks is nearly constant[2]. A similar effect was reported by *Birch* [1960], who found from studies of compressional wave velocity in rocks that, when the pressure surrounding jacketed rocks is raised, velocity increases more rapidly than for unjacketed rocks. Also, variations in velocity with direction of propagation, which often are present at low confining pressure, are generally found to disappear or diminish greatly when the confining pressure is raised.

These and other departures from the behavior of conventional materials, such as polycrystalline metals and glass, are usually explained by saying that 'solid rock' is not solid at all but contains many small fissures, created presumably upon removal of the high pressure and temperature present at formation. These flaws in the rock structure decrease the apparent moduli until reapplication of pressure to jacketed samples closes the flaws and makes the rock appear stiffer. When the rock sample is not jacketed, the fluid is free to enter most of the pores, the cracks have little tendency to close, and measurements reflect the deformation of the rock grains, which are nearly perfectly elastic. The presence of flaws and their closing under pressure, as first suggested by *Adams and Williamson* [1923] and now generally accepted, thus seems to offer a qualitative explanation for the large initial change in the elastic moduli with pressure.

In the analysis presented in this paper, we have evaluated theoretically the effect of cracks on deformation of rock by making use of a mathematical model in which rock is portrayed as an elastic isotropic material containing randomly oriented narrow cracks. At first glance, the use of an isotropic material as a model would not seem acceptable because of the notoriously anisotropic behavior of rock at low pressures. However, the anisotropic behavior appears to be due more to the effect of cracks themselves than to anisotropy in the crystalline framework of the rock. *Birch* [1961]

[1] Contribution 1553 from the Woods Hole Oceanographic Institution.

[2] Actually, compressibility of single crystals and other nonporous solids changes slightly with pressure, by a few per cent in 10 kb, but we ignore this effect here.

found in the measurement of sonic velocity in mutually perpendicular directions that raising the confining pressure on jacketed samples of even highly 'anisotropic' material such as slate renders them nearly isotropic to the passage of sound waves. To be sure, one property —compressive strength—has been shown by *Donath* [1961] for a slate to depend on orientation of the sample for any confining pressure. However, *Walsh and Brace* [1964] show that this behavior can be explained by assuming that the crystalline framework of the rock is elastic and isotropic but that the distribution of cracks is anisotropic; i.e., cracks parallel to the plane of anisotropy are longer than those in other orientations. Thus, anisotropy is generally caused by anisotropic crack distribution, and the assumption in our analysis that the material between cracks is isotropic and elastic appears justified.

Analysis

General. An expression for the effective compressibility β_{eff} of a porous elastic material can be derived in terms of the compressibility β of the solid material and the rate of change of porosity η with external pressure p. Consider a body whose external boundaries enclose a volume V_0 containing cavities of total volume V_c. *Chree* [1892] shows, using the reciprocal theorem, that the decrease in volume of the solid material, $\Delta(V_0-V_c)$, due to an increase in pressure Δp is given by $\beta\ \Delta p\ V_0$. In differential form,

$$d(V_0 - V_c) = \beta V_0 dp \qquad (1)$$

Differentiating and rearranging gives

$$(1/V_0)(dV_0/dp) = \beta + (1/V_0)(dV_c/dp) \qquad (2)$$

The first term in (2) is the definition of effective compressibility, β_{eff}. The last term, for strains small enough so that V_0 may be considered constant, is the rate of decrease of porosity with external pressure. Using the convention that an increase in porosity is positive, we have

$$\beta_{eff} = \beta - d\eta/dp \qquad (3)$$

Equation 3, based only on the requirement that the crystalline framework of the rock be linearly elastic, is independent of the shape or total volume of the pores. Experimental verification of (3) is not currently possible because no experiments are reported in which both the compressibility and the porosity have been measured as a function of pressure. It is interesting to note that β_{eff} should depend on the rate of change of porosity rather than on porosity directly. For some pore shapes (e.g., spherical pores), it can be shown that $d\eta/dp$ is a function of η, and in such cases elastic moduli and porosity are uniquely related. In general, however, η and $d\eta/dp$ are unrelated, and no correlation between the elastic properties and porosities of a suite of samples of the same rock should be expected

Spherical pores. The simple case of a body containing spherical cavities serves to illustrate (3). Consider the body to be separated into irregularly shaped regions, each containing one pore and each having the same porosity. Assume that the distance between pores is great enough so that the stress at the boundary of each region is very nearly the same as that at the boundary of the body. For the purpose of finding the change in the volume v_c of a small cavity due to the application of pressure p at the boundary of the region, the regime may be approximated by a sphere with volume v equal to that of the region. The tangential stress $\sigma_{\theta\theta}$ at the boundary of the cavity is found from Lamé's solution for thick-walled spheres [see *Timoshenko and Goodier*, 1951, p. 59] and can be expressed in the form

$$\sigma_{\theta\theta} = -(3p/2)(1 - v_c/v)^{-1} \qquad (4)$$

The tangential strain $e_{\theta\theta}$ $(1 - \nu)$ $(\sigma_{\theta\theta}/E)$, and the volumetric strain $\Delta\ v_c/v_c$ is $3e_{\theta\theta}$. Thus, the fractional change in the volume of the cavity for a differential change in pressure is

$$\frac{dv_c}{v_c} = -\frac{9}{2}\frac{dp}{E}\frac{1-\nu}{1-v_c/v}$$

and the rate of change of porosity with pressure is

$$\frac{d\eta}{dp} = \frac{dv_c}{v_0\,dp} = -\frac{9(1-\nu)}{2E}\frac{\eta}{1-\eta} \qquad (5)$$

Combining (5), (3), and the identity $\beta = 3(1 - 2\nu)/E$ results in

$$\beta_{eff} = \beta\left(1 + \frac{3}{2}\frac{1-\nu}{1-2\nu}\frac{\eta}{1-\eta}\right) \qquad (6)$$

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Equation 6 is identical with results obtained by *MacKenzie* [1950], *Eshelby* [1957], and *Hashin* [1959], each using different methods but relying on elastic theory for infinitesimal strains. It is strictly valid only for small porosity, although *Hashin* [1959] has considered finite concentration of cavities.

Duckworth et al. [1950], *Coble and Kingery* [1956], and *Hasselman and Fulrath* [1964], in tests on samples of porcelain, alumina, and glass, in which the porosity occurred as nearly spherical cavities, found excellent agreement between predicted and experimental values of elastic moduli. On the other hand, experimental values of moduli when no precautions are made to ensure spherical pores in the samples (e.g., *Knudsen* [1962] for alumina) were found to be lower than those predicted by the sperical-pore theory. Similarly, sonic velocity measurements on a suite of sandstones of varying porosities by *Shimozuru et al.* [1957] indicated a definite decrease in modulus with increasing porosity, although the scatter was considerable. The measured sound velocity for a suite of low-porosity granites by *Ide* [1936], however, bore virtually no relation to the porosities of the samples. The explanation lies in the likelihood that the sandstone and granite samples contained cracks which influenced the moduli without affecting the porosity significantly. Indeed, the porosity of the granite samples was probably due almost entirely to narrow cracks, and no correlation between elastic behavior and porosity should be expected.

Narrow cracks. The effect of a random distribution of narrow cracks upon the compressibility of an elastic body is treated as in the spherical pore case, by first dividing the body into regions, each of which contains one crack. One difficulty not present in the spherical-pore case immediately arises—that of deciding the boundary conditions to be applied to each region. If the stress at the region boundary is assumed to be equal to the applied hydrostatic pressure, the deformation of the boundary, although having a symmetry corresponding to that of the crack, is not uniform. Thus, continuity of displacements cannot be maintained at the boundary of two regions containing cracks of different orientations. Similarly, if the strain at the boundary of a region is assumed to be uniform and equal to that of the body, the stress on the region boundary is not uniform, and equilibrium between adjacent regions is, in general, not satisfied. This same problem has occurred in computing the elastic constants for polycrystalline aggregates from the moduli of the components. *Voigt* [1928] suggested averaging the elastic moduli, which is equivalent in the present problem to assuming a uniform strain on the boundary of each region, whereas *Reuss* [1929] averaged the compliances, which corresponds to the assumption of uniform stress on the region boundary. *Hill* [1952] showed that the Voigt and Reuss methods are upper and lower bounds on the moduli; the separation of the bounds depends on the degree of anistropy of the components. The Voigt-Reuss bounds are considered adequate for the present analysis, since the anistropy of individual regions, which depends on the ratio of crack length to region size, is small for the assumed low concentration of cracks.

The shape of the crack must be specified in order to calculate its effect upon the compressibility of a region. We shall investigate three possibilities—the penny-shaped crack, the elliptical crack in plane stress, and the elliptical crack in plane strain. The results will be found to be quite similar.

The compressibility of a region is found somewhat more conveniently by using, instead of (3), equation 2 rearranged as follows:

$$\beta_{eff} = (\beta V_0 p \, dp + p \, dV_c)/V_0 p \, dp$$

or, for a single region,

$$\beta_{eff} = (\beta v_0 p \, dp + p \, dv_c)/v_0 p \, dp \qquad (7)$$

The first term in the numerator on the right-hand side of (7) is the strain energy associated with the region if there is no crack, and the second term is the increase in strain energy dw_c due to the crack. This second term is of interest in the Griffith fracture theory and therefore has been calculated for several cases—the penny-shaped crack by *Sack* [1946] and the elliptical crack in plane stress and plane strain by *Griffith* [1920]. For very narrow cracks of half-length and half-depth c, with uniform pressure p on the region boundary, the expressions are

$$dw_c = \beta \frac{16(1 - \nu^2)c^3}{9(1 - 2\nu)} p \, dp \quad \text{(penny-shaped)}$$

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$$dw_c = \beta \frac{4\pi(1-\nu^2)c^3}{3(1-2\nu)} p\, dp \quad \text{(plane strain)}$$

$$dw_c = \beta \frac{4\pi c^3}{3(1-2\nu)} p\, dp \quad \text{(plane stress)} \qquad (8)$$

The total strain energy dW_c due to all cracks is found by summing the values dw_c in (8) over the total number of regions N. For the penny-shaped crack, for example,

$$dW_c = \beta \sum_N \frac{16(1-\nu^2)c^3}{9(1-2\nu)} p\, dp \qquad (9)$$

Similarly, the expression for β_{eff} is found from (9) and (7), where $dW_c = p\, dV_c$:

$$\beta_{eff} = \beta\left[1 + \frac{16(1-\nu^2)}{9(1-2\nu)V_0} \sum_N c^3\right] \qquad (10)$$

If we define an average crack length $\bar{c}$, where $N\bar{c}^3 = \Sigma_N c^3$ and an average region volume $\bar{v}$, where $N\bar{v} = V_0$, the effective compressibility for the Reuss assumption is found to be

$$\beta_{eff} = \beta\left(1 + \frac{16}{9}\frac{1-\nu^2}{1-2\nu}\frac{\bar{c}^3}{\bar{v}}\right) \quad \text{(penny-shaped)}$$

$$\beta_{eff} = \beta\left(1 + \frac{4\pi}{3}\frac{(1-\nu^2)}{1-2\nu}\frac{\bar{c}^3}{\bar{v}}\right) \quad \text{(plane strain)}$$

$$\beta_{eff} = \beta\left(1 + \frac{4\pi}{3(1-2\nu)}\frac{\bar{c}^3}{\bar{v}}\right) \quad \text{(plane stress)} \qquad (11)$$

The expressions for β_{eff} derived from the Voigt assumption require a somewhat longer calculation, since expressions for the strain energy of a cracked solid with a uniform-strain boundary condition are not available in the literature and must be derived. From an analysis in the appendix, the effective compressibility using the Voigt model approaches that of the Reuss model for very small crack concentrations. The relationship between the Voigt and Reuss values of the effective compressibility (equation A-10) is

$$(\beta_{eff})_V = (\beta_{eff})_R - \beta \sum_N \frac{2A^2(v/V)}{9(1-2\nu)(1+\nu)} \qquad (12)$$

where A, defined by equation A-1 for various crack types, is a parameter proportional to c^3/v. Equation 12 shows that the Voigt value for compressibility is less than the Reuss value and that the difference between the two decreases with decreasing concentration as predicted by *Hill* [1952]. However, since the crack concentration has been assumed to be small, the summation term is of second order and, to the order of approximation of this analysis, the Voigt and Reuss values should be assumed equal.

Crack closure. The compressibility of a region after the crack has closed is the same as that of uncracked material, and it remains essentially constant with further changes in pressure. Thus the rock gradually becomes stiffer as cracks close under increased confining pressure. For rocks where the entire porosity is caused by cracks, the compressibility eventually decreases to a nearly constant value equal to that for an uncracked sample, or, equivalently, that of unjacketed samples where the fluid can freely enter the pores. For rocks like sandstone, where cracks are responsible for a relatively small proportion of the total porosity, the compressibility of the sample again decreases to a nearly constant value. However, the final compressibility is not that of a 'solid' sandstone but that of a porous sandstone, since porosity other than that caused by rather narrow cracks is virtually impossible to remove even with very high confining pressures. This can be illustrated by finding the pressure required to close a penny-shaped crack with major axis $2\bar{c}$ and a ratio of minor to major axis α. The volume of the ellipsoidal cavity is $(4/3)\pi\bar{c}^3\alpha$, so the rate of change of porosity with pressure, where the region volume is $\bar{v}$, is

$$d\eta/dp = (4\bar{c}^3/3\bar{v})\, d\alpha/dp \qquad (13)$$

The rate of change of porosity can also be found by equating (3) and (11) for the penny-shaped crack:

$$d\eta/dp = -16\beta(1-\nu^2)\bar{c}^3/9(1-2\nu)\bar{v} \qquad (14)$$

Equations 13 and 14 are equated, and the resulting equation is integrated to give the pressure p_c required to reduce α to zero.

$$p_c = \pi E\alpha/4(1-\nu^2) \qquad (15)$$

Deriving similar expressions for the elliptical crack in plane stress and plane strain gives similar results: to an order-of-magnitude approximation, the pressure required to close a

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crack is $E\alpha$. Now consider a porous sandstone composed of quartz grains, for which Young's modulus is about 1000 kb. The pressure required to close cavities with aspect ratios of one-tenth is 100 kb, a pressure beyond the capabilities of most apparatus. Some cavities are likely to be more nearly spherical and to require even a higher pressure. Thus, the only cavities which close under pressure must be very narrow cracks. The experiments discussed by *Birch* [1961], for example, where application of several kilobars served to close the cracks, indicate that the cracks had an aspect ratio of the order of 1 to 1000.

The closing of cracks under hydrostatic pressure provides a useful method for determining the porosity of rock for which all porosity is in the form of cracks. For such rocks, the porosity, which is less than 1% or so and difficult to measure, can be found from the plot of volmetric strain versus pressure taken during a compressibility test. First, consider (2) integrated to give an expression for the change in porosity $\Delta\eta$.

$$-\Delta V_c/V_0 = \Delta\eta = \Delta V_0/V_0 - \beta p \qquad (16)$$

The total change in porosity when all cracks have closed is equal to the initial porosity η_0. If we assume that all cracks are closed at a pressure p_1 corresponding to a volumetric strain $(\Delta V/V)_1$, (16) becomes

$$\eta_0 = (\Delta V_0/V_0)_1 - \beta p_1 \qquad (17)$$

Reference to Figure 1, a plot of pressure versus volumetric strain, shows that the porosity η_0 in (17) is simply the intercept on the volumetric strain axis of the straight-line portion of the curve extrapolated back to zero pressure. The change in porosity $\Delta\eta$ due to application of pressure p_2 less than that required to close all the cracks can be shown by using (16) to be the intercept on the volumetric strain axis of a line dropped at an angle equal to the compressibility. It is necessary, of course, that the slope of the straight-line portion (see footnote 2) of the volumetric strain-pressure curve correspond to the compressibility of a sample with zero porosity for this graphical method to be valid. This can be checked by verifying that the slope of the straight-line portion is equal to the slope of the plot for an unjacketed sample where fluid is free to enter all pores.

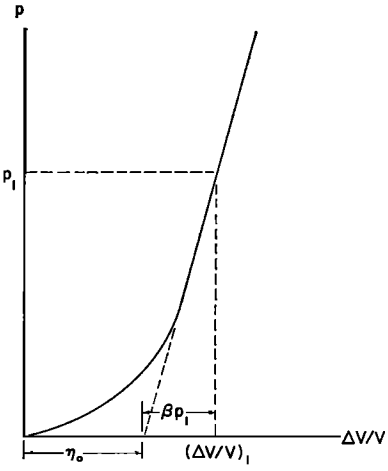


Fig. 1. Graphical determination of porosity (see text).

Brace [1965] compared porosity, determined graphically as described above, with that measured directly for granites from Westerly, R. I., and Stone Mountain, Ga. The graphical method, applied to three mutually perpendicular samples cored from a single large block of uniform material, gave the quantity called 'crack porosity,' η_c, with an uncertainty of about 0.0005 (Table 1). Total porosity, η, of the two granite specimens was measured by standard methods [*Dreyer*, 1958], again with an uncertainty of about 0.001 (Table 1). For the Stone Mountain granite, η_c was equal to η to within the errors of the measurements. For Westerly granite, however, η_c was much smaller than η. By comparison of intrinsic compressibility, calculated from mineral content, with observed com-

TABLE 1. Porosity of Two Granites*

	From Compressibility Data			Measured
Sample	η_c ± 0.0005	η_p ± 0.003	η	η ± 0.001
Granite Stone Mt., Ga.	0.0035	0	0.0035	0.003
Granite Westerly, R. I.	0.0020	0.012	0.014	0.011

* After *Brace* [1965].

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pressibility at pressures of 5 to 9 kb, the reason for this discrepancy was clear. Evidently, a rather large amount of porosity persisted at these high pressures in Westerly, whereas Stone Mountain granite had none. An estimate of this 'pore porosity,' η_p, from (6), shown in Table 1, appeared to account for the difference noted above at least to within the uncertainty in measurement of η_p (0.003).

Discussion

In the preceding analysis, an equation was derived which shows that the difference between the compressibilities of a porous body and a solid body of the same composition is equal to the rate of change of porosity with externally applied pressure for the porous body. This relationship is independent of the shape or concentration of the pores and relies only on the assumption of linear elasticity and small strains. Expressions for the compressibility of porous bodies are found by assuming a pore shape and concentration for which the rate of change of porosity can be determined theoretically. Two cases were considered—low concentrations of spherical pores and narrow cracks.

A comparison of spherical pores and narrow cracks illustrates the effect of pore geometry upon compressibility. If we assume that all spherical pores have the same diameter $\bar{c}$, the compressibility (6) for low porosity is

$$\beta_{eff} = \beta\left(1 + 2\pi \frac{1 - \nu}{1 - 2\nu} \frac{\bar{c}^3}{\bar{v}}\right) \qquad (18)$$

Equation 18 is identical with the expressions derived for narrow cracks (11), with the exception of the magnitude of the coefficient of the last term. Evaluating the coefficient for various values of Poisson's ratio shows that spherical holes are some two to three times more effective than cracks of the same radius in increasing compressibility. Thus we find that the effect upon compressibility of a certain concentration of narrow cracks equals the effect of the same number of spherical pores with a diameter roughly two-thirds that of the cracks. The porosity of the two samples differs enormously, of course. Also, the analysis shows that compressibility is independent of α, the aspect ratio of a crack, so that a crack which is about to close is as effective as a fully open crack of the same length.

The compressibility is strongly affected by the presence of a few long cracks. As shown in (11), in which crack length appears to the third power, one crack ten units long has the same effect as 1000 cracks one unit long. The strong influence of the longer cracks accounts for the apparent anisotropy at low pressure of seemingly isotropic rocks because of the improbability of a small collection of long cracks having a random character.

An interpretation of some interesting measurements of linear compressibility (linear strain per unit pressure) as a function of pressure by *Brace* [1965] summarizes the effects of cracks and crack closure derived above. Linear compressibilities measured parallel ($||$) and perpendicular ($\perp$) to foliation of a gneiss sample were found to have the unusual pressure dependence shown in Figure 2, in which $\beta_\perp$ is first larger, then smaller, and finally again larger than $\beta_\parallel$ as the pressure is increased. The rock contained about 10% mica in the form of long flakes with a strong preferred orientation; the remainder of the rock was quartz and feldspar with virtually random distribution of crystallographic axes. If we assume that grain boundaries form the sites of cracks, here is a rock with a few highly oriented long cracks among a large number of randomly oriented shorter cracks. In Figure 2, at the lowest pressure, $\beta_\perp$ is higher because the cracks perpendicular to that direction are longer. At the highest pressures all cracks have presumably closed, but $\beta_\perp$ is still higher than $\beta_\parallel$ because mica is more compliant normal than parallel to (001), the plane of anisotropy. Between the highest and lowest pressures, the two curves cross. Here we see the effect of crack shape upon closure. The longer cracks tend to close at a lower pressure than the shorter, causing $\beta_\perp$ to fall to its intrinsic value at a lower pressure than $\beta_\parallel$.

Appendix

Consider a region of volume v with uniform strain e on the boundary produced by stress R, S, and T, where R and T are in the plane of the crack and S is normal to the crack. The strain energy of the region is considered to be composed of two parts—the amount dw_s if there is no crack plus an amount dw_c attributed to the

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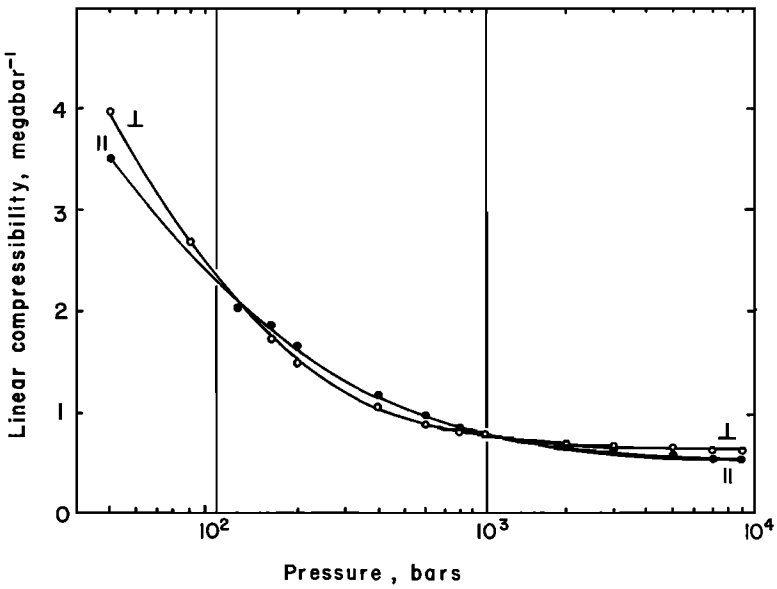


Fig. 2. Linear compressibility of Torrington gneiss parallel (||) and perpendicular (⊥) to foliation (after *Brace* [1964]).

presence of the crack. *Sack* [1946], for the penny-shaped crack, and *Griffith* [1920], for the crack in plane stress and plane strain, show that the strain energy, dw_c, associated with the crack is independent of the values of R and T and is given by

$$dw_c = AvS\ dS/E$$

where

$$A = 16(1 - \nu^2)c^3/3v \quad \text{penny-shaped}$$

$$A = 4\pi(1 - \nu^2)c^3/v \quad \text{plane strain}$$

$$A = 4\pi c^3/v \quad \text{plane stress} \qquad \text{(A-1)}$$

The strain energy dw_s for solid region of volume v is

$$dw_s = \{[R - \nu(S + T)]\ dR + [S - \nu(R + T)]\ dS + [T - \nu(S + R)]\ dT\}v/E \qquad \text{(A-2)}$$

The total strain energy for a region is the sum of (A-1) and (A-2). The average strain in each principal direction is given by

$$v^{-1}\ \partial w/\partial R \qquad v^{-1}\ \partial w/\partial S \qquad v^{-1}\ \partial w/\partial T$$

or

$$Ee = R - \nu(S + T)$$
$$Ee = S(1 + A) - \nu(R + T)$$
$$Ee = T - \nu(R + S) \qquad \text{(A-3)}$$

The stresses can be found in terms of the boundary strain e by solving (A-3).

$$R = T = \frac{3e}{\beta}\frac{1 + A/(1 + \nu)}{1 + A(1 - \nu)/(1 + \nu)(1 - 2\nu)}$$
$$S = \frac{3e}{\beta}\left[1 + \frac{A(1 - \nu)}{(1 + \nu)(1 - 2\nu)}\right]^{-1} \qquad \text{(A-4)}$$

The strain energy for a region is found by substituting (A-4) in the sum of (A-2) and (A-1).

$$dw = \frac{9Ede}{\beta} \cdot \frac{1 + \dfrac{A(5 - 7\nu)}{3(1 + \nu)(1 - 2\nu)} + \dfrac{2A^2(1 - \nu)}{3(1 + \nu)^2(1 - 2\nu)}}{[1 + A(1 - \nu)/(1 + \nu)(1 - 2\nu)]^2} v \qquad \text{(A-5)}$$

The total strain energy is found by summing (A-5) over all N regions.

$$dW = \frac{9ede}{\beta} \cdot \sum_N \frac{1 + \dfrac{A(5 - 7\nu)}{3(1 + \nu)(1 - 2\nu)} + \dfrac{2A^2(1 - \nu)}{3(1 + \nu)^2(1 - 2\nu)}}{[1 + A(1 - \nu)/(1 + \nu)(1 - 2\nu)]^2} \qquad \text{(A-6)}$$

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The total strain energy is work done by the pressure on the total volume V, or

$$dW = \beta_{eff}(Vp\ dp) \qquad \text{(A-7)}$$

Combining (A-6) and (A-7), where $e = \beta_{eff}p/3$, gives an expression for β_{eff}.

$$\frac{1}{\beta_{eff}} = \frac{1}{\beta} \cdot \sum_N \frac{1+\frac{A(5-7\nu)}{3(1+\nu)(1-2\nu)}+\frac{2A^2(1-\nu)}{3(1+\nu)^2(1-2\nu)}}{[1+A(1-\nu)/(1+\nu)(1-2\nu)]^2}\frac{v}{V} \qquad \text{(A-8)}$$

An approximate expression for β_{eff}, correct to the order of approximation of the analysis, can be found by expanding (A-8).

$$\frac{1}{\beta_{eff}} = \frac{1}{\beta}\left[\sum_N \frac{v}{V}\left(1 - \frac{A}{3(1-2\nu)} + \frac{A^2}{3(1+\nu)(1-2\nu)^2} + \cdots\right)\right]$$

or

$$\beta_{eff} = \beta\left(1 + \sum_N \frac{A(v/V)}{3(1-2\nu)} - \sum_N \frac{2A^2(v/V)}{9(1-2\nu)(1+\nu)} = \cdots\right) \qquad \text{(A-9)}$$

It can be shown that the first two terms on the right-hand side of (A-9) give the effective compressibility for the Reuss assumption. Thus, the Voigt and Reuss values of the compressibility are related by

$$(\beta_{eff})_V = (\beta_{eff})_R - \beta \sum_N \frac{2A^2(v/V)}{9(1-2\nu)(1+\nu)} \qquad \text{(A-10)}$$

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